

STOCK MOVEMENT CONTEXT DASHBOARD

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GitHub Repository: https://github.com/04Shreya13/Stock_Movement_Context_Dashboard

Data Source Link: <https://huggingface.co/datasets/TheFinAI/Herculean>

Our Vision

An investor-relations analyst at a large investment firm must determine the cause in sharp movements of eight large-cap companies in their portfolio before talking to investors about specific movements to make. Specifically, they decide to review the news and filings of each company as they may point to company-specific developments. The analyst selects a company, sees the days when the daily price range was unusually wide, and opens one day to compare the news highlights and the latest 10-K or 10-Q discussions. He repeats this process with the other companies. The output is a shortlist of suggested company-specific movements for human follow-up.

ESTABLISHING THE DATA FIT

1a. Source and Scope:

Herculean (TheFinAI/Herculean on Hugging Face) is an offline benchmark built to test LLM agents on financial-analysis tasks: trading, hedging, report generation, report evaluation, and XBRL filing auditing. Its core data is three tables covering eight US mega-cap stocks - prices, news and filings.

Table	One Row =	Rows
prices	one company on one trading day	2,656
news	one company on one calendar day (an LLM-written highlights summary)	3,888
filings	one 10-K or 10-Q for one company	73

What we opened?

Total file size: 1.48 GB

We loaded the full prices (2,656 rows), news (3,888 rows), and filings (73 rows) tables from the dataset's Parquet files using pandas. The tables are small enough that we did not need a sample. Covers AAPL (Apple), ADBE (Adobe), AMZN (Amazon), GOOGL (Google), META, MSFT (Microsoft), NVDA (Nvidia), and TSLA (Tesla) for the below mentioned dates:

prices	2024-12-02	to	2026-03-31
news	2024-12-01	to	2026-03-31
filings	2024-01-16	to	2026-03-25

What the data can support:

- We can support a dashboard that identifies and compares stock price movements across the eight companies in the dataset. Using the prices, news, and filings tables, we can analyze price changes and display relevant news and company filing information around these movements.
- Flagging each company's unusually large daily price ranges and comparing them across the 8 companies.
- Showing news highlights and the latest 10-K or 10-Q around each flagged day.
- Marking days when several companies were flagged together, as a partial check for market-wide moves.

Missing:

The available data may not provide enough information of all factors that may affect a stock price movements, including broader market conditions, macroeconomic events, industry events, etc. Therefore, we can provide potential factors that relate to the price movements, but cannot establish causations.

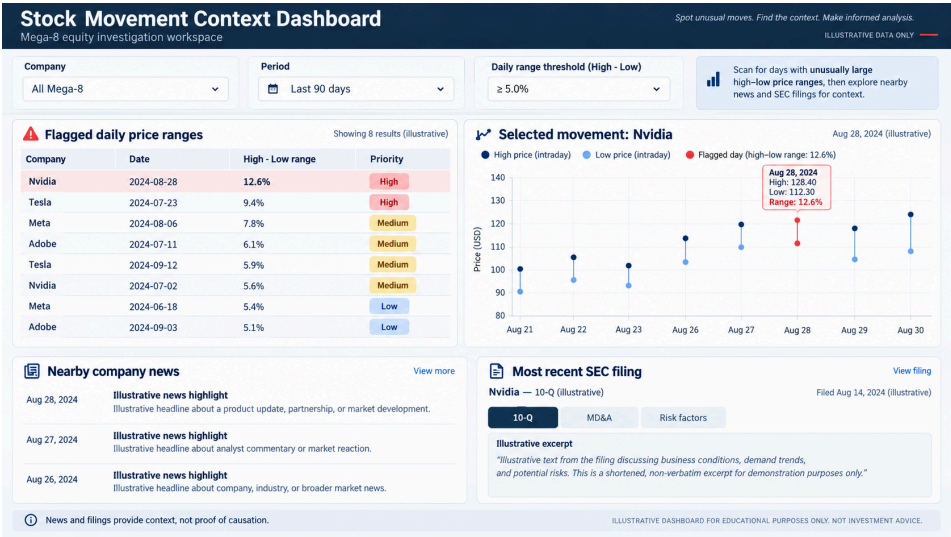
Relevant columns:

Source Table	Column Name	What it contains	How it will be used
Prices, Filings, News	Symbol	The ticker symbol for each company	Connect a company’s daily stock-price range with its related filings and news.
Prices, Filings, News	Date	The date associated with a price record, filing, or news item	Identify the day a large price range occurred and review company information from that time period.
Prices	High, Low	The highest price reached by a company’s stock during one trading day	Identify days when the stock reached an unusually high or low price.
News	Highlights	Key highlights from company-related news	Help explain what company events or announcements may have occurred around a large daily price range.
Filings	MDA Content	Management discussion and analysis text	Show management’s explanation of business performance and possible concerns.

Example of a small pandas exploration - hypothetical results:

In the 2,656-row prices table covering 8 companies (Dec 2024 to Mar 2026), 136 trading days are flagged as big moves, meaning each company's top 5% of days by daily range (high – low, divided by close), which is 17 days per company. All 136 flagged days have a matching news row. These results describe this sample of 8 stocks over this period, not all stocks or time periods.

Dashboard Sketch



How the sketch connects to our data

1. Dropdown filters: Choose a company, time period, and daily price-range threshold. Company filtering uses symbols and time filtering uses date. The threshold is based on the daily range calculated as high - low.

2. Flagged daily price-ranges table: Group the prices table by symbol and compare daily price ranges. The dashboard calculates high - low to flag dates with unusually large intraday movement. The table shows the company, date, high-low range, and priority for human review.

3. Selected movement chart: For the selected company and date, show the day's high and low prices and compare them with nearby trading days. This helps the analyst see whether the flagged range was isolated or part of a larger pattern.

4. Nearby company-news panel: Display highlights from the news table for the selected company around the flagged date, such as the day itself and the previous seven days. These items may provide useful context, but they do not prove what caused the movement.

5. SEC filing panel: Display the most recent available company filing using document_type, mda_content, and risk_content from the filings table. This lets the analyst review management discussion and company-reported risks that may be relevant.

Initial scope: The dashboard will cover the eight Mega-8 companies, use high - low to identify unusually large daily ranges, show nearby news and the most recent filing, and support one analyst-review decision. It will not predict future stock prices, provide investment recommendations, or claim that news or filings caused a movement.

